- **Name:** Sumit Raj
- **Role:** Software Developer | Data Scientist | AI/ML Engineer
- **Location:** India

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EDUCATION

M.Tech in Applied AI & ML (Pursuing)
Visvesvaraya National Institute of Technology (VNIT), Nagpur
2024 – Present

B.E. in Computer Science | CGPA: 7.89/10
Dr. DY Patil Institute of Technology, Pune
July 2023

HSC – 69.86%
Gurukul Vidyalaya Muzaffarpur Islamia College, Darbhanga
Feb 2019

SSC – 75.40%
Gurukul Vidyalaya Muzaffarpur Islamia College, Darbhanga
Mar 2015

TECHNICAL SKILLS

Programming Languages

Python, Java, C++, C, R, SQL, JavaScript, TypeScript, HTML, CSS, MATLAB

Data Science & Machine Learning

Machine Learning, Deep Learning (CNN, RNN, LSTM), Data Analysis, Data Visualization, Feature Engineering, Statistical Modeling, NumPy, Pandas, Scikit-learn, TensorFlow, Keras, Matplotlib, Seaborn

Software Development

Full-Stack Development, REST APIs, OOP, Software Engineering, System Design, Git/GitHub

Databases

MySQL, SQLite, MongoDB, DBMS

Tools & Technologies

Google Cloud, MS Excel, Power BI, MLflow, Docker, VS Code, Jupyter Notebook, Colab

Core Skills

Competitive Programming, Algorithms, DSA, Problem Solving, Engineering Mathematics

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ACHIEVEMENTS

- 5-Star Java Programmer on HackerRank (Gold Badge in Problem Solving)
- 4-Star C++ Programmer on HackerRank
- 3-Star in Algorithms on HackerRank
- Solved **300+ coding problems** across multiple platforms
- Completed Google Cloud Fundamentals in Core Infrastructure
- Completed "Defend Against Threats with Microsoft 365"
- Completed "Threat Modeling Security Fundamentals"

PROFESSIONAL EXPERIENCE

Software Engineer Intern | Salesforce

Feb 2022 – May 2022

- Worked on Salesforce-based application development and workflow automation.
- Integrated bulk data pipelines and streamlined backend processes.
- Improved knowledge of Salesforce Administration and CRM systems.
- Gained hands-on experience in application development and cloud services.
- Built secure networking and data management modules.

PROJECTS

1. Real-Time Automated Military Security System (2025)

- Developed an automated security and surveillance system for military camps to overcome limitations of manual identity checks and static CCTV monitoring.
- Implemented **Haar Cascade-based face authentication** to verify authorized personnel and detect unknown individuals.
- Captured and sent real-time **email alerts** with the image, timestamp, and location of unknown individuals.
- Integrated **YOLOv8 object detection** to identify, classify, and track military vehicles with automatic alert generation.
- Enhanced real-time threat detection accuracy, minimized human dependency, and strengthened overall security and situational awareness in sensitive zones.

2. Brain Tumor Detection Using CNN (2025)

- Implemented a Convolutional Neural Network (CNN) for MRI-based brain tumor detection.
- Preprocessed MRI datasets with augmentation techniques.
- Achieved high classification accuracy using TensorFlow and Keras.
- Analyzed tumor features using deep learning visualization.

3. Mobile Botnet Detection (Jan 2023)

- Developed a botnet detection system for Android devices.
- Used Support Vector Machines (SVM) and SQLite for classification.
- Detected malicious iBots communicating with a botmaster via C&C server.

4. URL Shortener (May 2023)

- Built a full-stack URL-shortening web application.
- Implemented backend logic using JavaScript and database storage.
- Deployed link analytics and redirection features.

CODING PROFILES & CERTIFICATIONS

- **HackerRank:** 5-Star Java, 4-Star C++, Algorithm 3-Star
- **Google Cloud:** Cloud Fundamentals, Security Certifications
- **Microsoft:** Threat Modeling & Cloud Security

DECLARATION

I hereby declare that the above information is true to the best of my knowledge.